

Wind Driven Ocean Gyres Sverdrup and Munk Solutions

Fredrik Bergel'v

June 4, 2026

Introduction

Ocean circulation gyres are part of the atmospheric circulation.

- 1 Heating air at the equator → Hadley cells.
- 2 Coriolis effect deflects the air movement → Trade winds and westerlies
- 3 Enforce wind stress on the ocean surface → 90° Ekman transport.
- 4 Since f increases with latitude, net southward Ekman transport in the North Atlantic.
- 5 Western boundary current needed to close the circulation.



Figure: Wikimedia Commons, North Atlantic location map, accessed 2026

Introduction, Sverdrup

Introduction

Theory

Sverdrup

Munk

Method

Result

The Sverdrup
solution

The Munk
solution

Discussion

Conclusion

References

Harald Ulrik Sverdrup (1888-1957) formulated the net transport in the ocean interior as:

$$\boxed{-\beta \frac{\partial \Psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau}} \quad (1)$$

which gives the Sverdrup solution:

$$\begin{aligned} \int_x^{x_{\text{East}}} \frac{\partial \Psi}{\partial x} dx &= \int_x^{x_{\text{East}}} v H dx = \cancel{\Psi(x_{\text{East}}, y)} - \Psi(x, y) \\ \Rightarrow \Psi(x, y) &= -\frac{1}{\rho \beta} \int_x^{x_{\text{East}}} (\nabla \times \vec{\tau}) \end{aligned}$$

Introduction, Munk

Walter Heinrich Munk (1917-2019) extended the Sverdrup model by including meridional harmonic Newtonian viscosity (diffusion of momentum):

$$A \nabla^4 \Psi - \beta \frac{\partial \Psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau} \quad (2)$$

which gives the Munk solution:

$$\Psi(x, y) = \Psi_{\text{Sverdrup}}(x, y) \left(1 - e^{-\frac{x - x_{\text{West}}}{2\delta_m}} \left[\cos\left(\sqrt{3} \frac{x - x_{\text{West}}}{2\delta_m}\right) + \frac{1}{\sqrt{3}} \sin\left(\sqrt{3} \frac{x - x_{\text{West}}}{2\delta_m}\right) \right] \right)$$

$$\text{where } \delta_M \equiv \left(\frac{A}{\beta} \right)^{\frac{1}{3}}$$

Stress set as:

$$\tau = \rho_{\text{air}} C_D U_{10}^2 \sin \left(\frac{\pi y}{L_y} \right). \quad (3)$$

<i>Variable</i>	<i>Sverdrup model</i>	<i>Munk model</i>
Atlantic, L_x, L_y	7000 km, 7000 km	7000 km, 7000 km
Pacific, L_x, L_y	–	14000 km, 7000 km
Grid, $\Delta x, \Delta y$	35 km	35 km
Wind forcing, U_{10}	[4, 5, 6] m/s	5 m/s
Viscosity, A	0	$[3 \cdot 10^4, 10^5, 5 \cdot 10^5] \text{ m}^2/\text{s}$
$\delta_M / \Delta x$	–	[3.3, 4.9, 8.4]

Table: Comparison of parameters used in the Sverdrup and Munk models.

Result, The Sverdrup solution

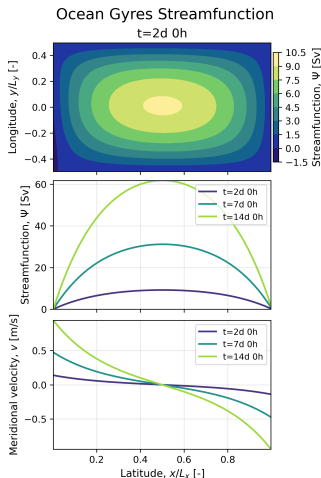


Figure: Streamfunction for the no-drag f -plane configuration. The circulation remains symmetric with no β -effect and without western intensification.

Result, The Sverdrup solution

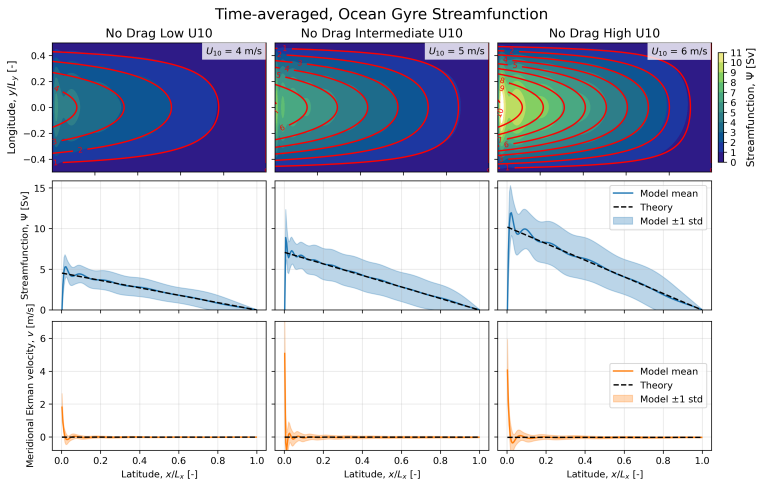


Figure: Streamfunction for the no-drag β -plane configuration. The red lines are the theoretical Sverdrup solution.

Result, The Munk solution

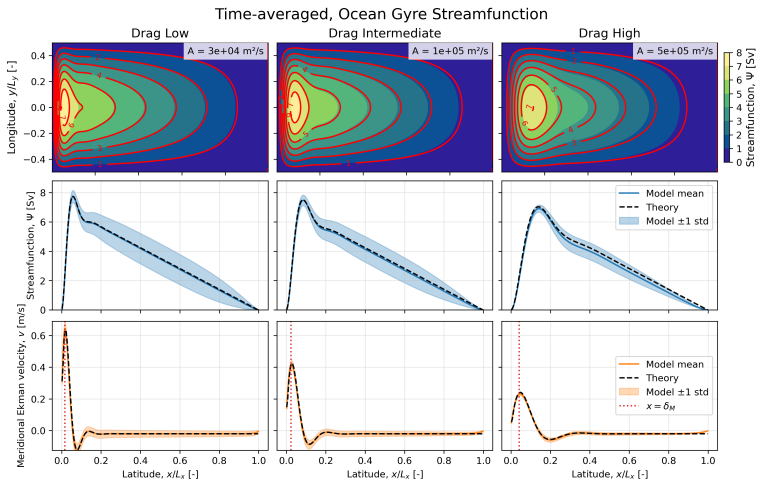


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Munk solution.

Result, The Munk solution

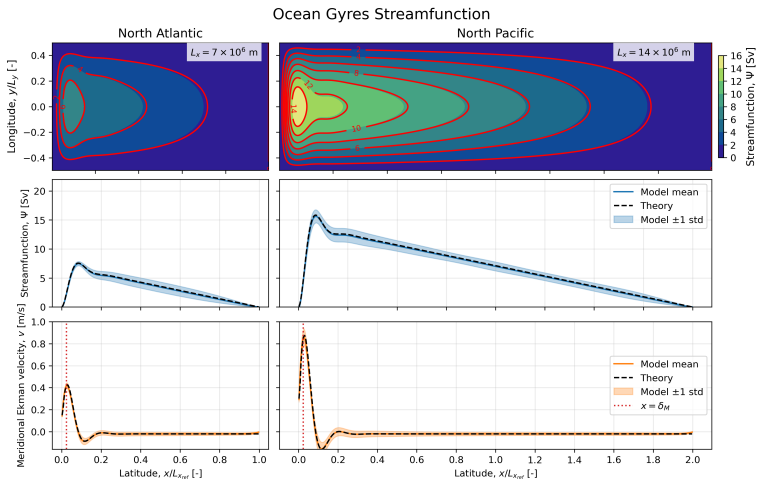


Figure: Streamfunction for the drag β -plane configuration with varying domain size. The red lines are the theoretical Sverdrup solution.

Result, The Munk solution

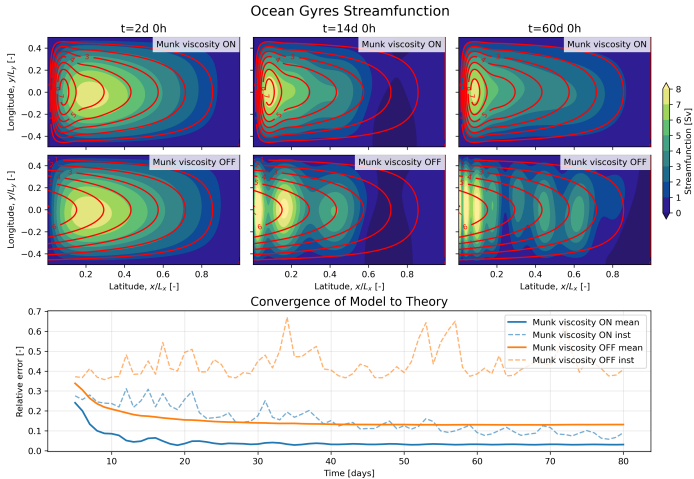


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Sverdrup and Munk solutions respectively.

Both numerical models show western boundary currents, but only Munk theory does. Why?

- The Sverdrup theory is linear, whereas the Munk theory is non-linear, and frictional force is opposite sign to the motion itself.
- The Munk viscosity creates a boundary layer where the frictional force can take care of mass conservation.
- Friction balances continuous input of energy. *Realistic?*
- Gives western boundary flow in right magnitude to real Gulf Stream ($\sim 2.5 \text{ m s}^{-1}$).

$$A \nabla^4 \psi - \beta \frac{\partial \psi}{\partial x} = \frac{1}{\rho} \nabla \times \vec{\tau}$$

Three important conclusions:

- 1 Both numerical models showed western intensification, explained by Rossby wave propagation. But continuous energy input need dissipation to converge with time.
- 2 Only Munk theory showed a western boundary current, explained by frictional layer, although both numerical models showed return flow.
- 3 A larger domain gave same interior meridional velocity, which resulted in a stronger western boundary current.

References

-  Döös, Kristofer, Peter Lundberg, and Aitor Aldama Campino (2022). *Basic Numerical Methods in Meteorology and Oceanography*. Stockholm, Sweden: Stockholm University Press. ISBN: 978-91-7635-172-7. DOI: 10.16993/bbs.
-  Ekman, V. Walfrid (May 1905). "On the Influence of the Earth's Rotation on Ocean-Currents.". In: *Arkiv för matematik, astronomi och fysik* 2.11, pp. 1–52.
-  Munk, Walter H. (1950). "On the wind-driven ocean circulation". In: *Journal of Meteorology* 7.2, pp. 79–93.
-  Steele, John H. (2008). *Encyclopedia of Ocean Sciences*. 2nd ed. Amsterdam: Academic Press. ISBN: 9780123744739.
-  Stommel, Henry (1948). "The westward intensification of wind-driven ocean currents". In: *Transactions, American Geophysical Union* 29.2, pp. 202–206.
-  Sverdrup, H. U. (1947). "Period increase of ocean swell". In: *Eos, Transactions American Geophysical Union* 28.3, pp. 407–417.
-  Vallis, G. K. (2006). *Atmospheric and Oceanic Fluid Dynamics*. Cambridge, U.K.: Cambridge University Press, p. 745.

Extra figures

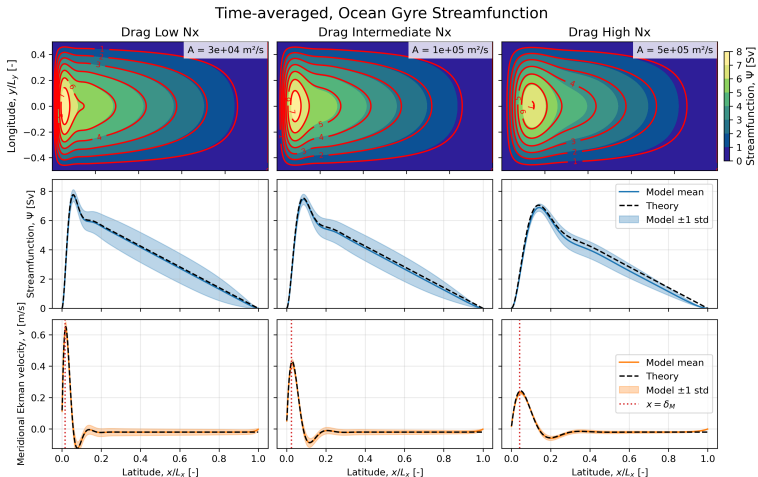


Figure: Streamfunction for the Laplace-viscous β -plane configuration. The red lines are the theoretical Munk solution. This figure has three times as high resolution (i.e. $\Delta x = \Delta y = 11.7 \text{ km} \Rightarrow \frac{\delta_M}{\Delta x} \approx [10, 15, 25]$)